

Segment Anything for Cell Tracking

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Goal

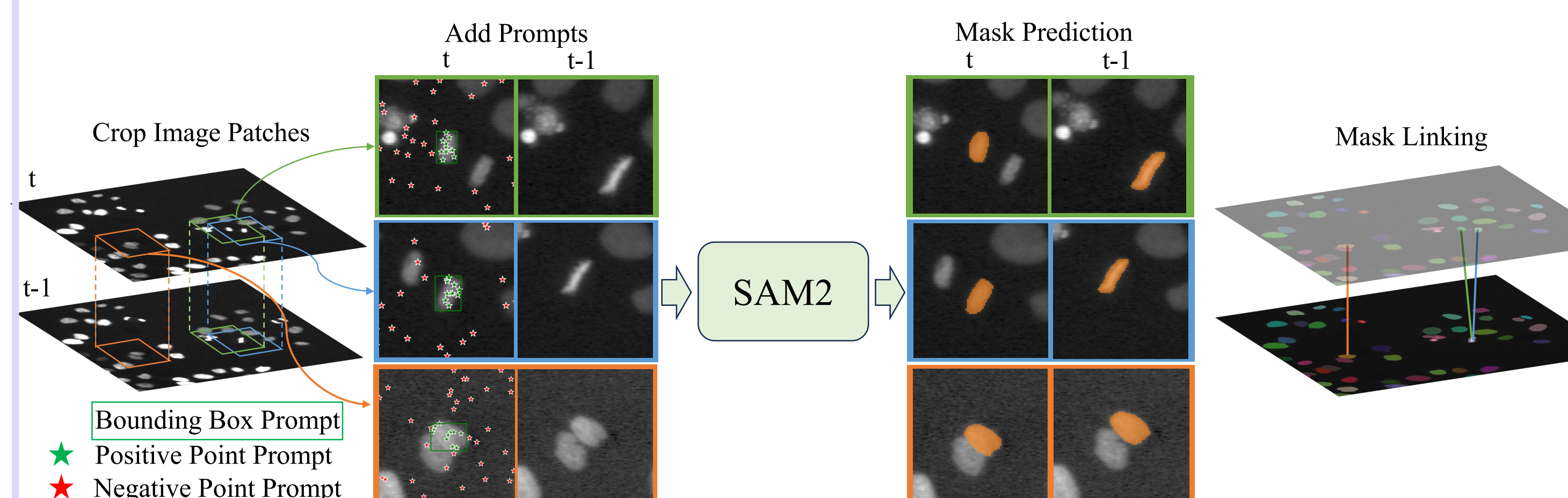
- Zero-shot, annotation-free pipeline for 2D & 3D cell tracking using Segment Anything 2 (SAM2) [1] and SAM-Med3D [2]:
 - Linking-only mode** when segmentation is provided.
 - Segmentation + tracking mode** for large 3D datasets and selected subsets of cells.
- Interactive user interface for mask selection and correction.

Introduction

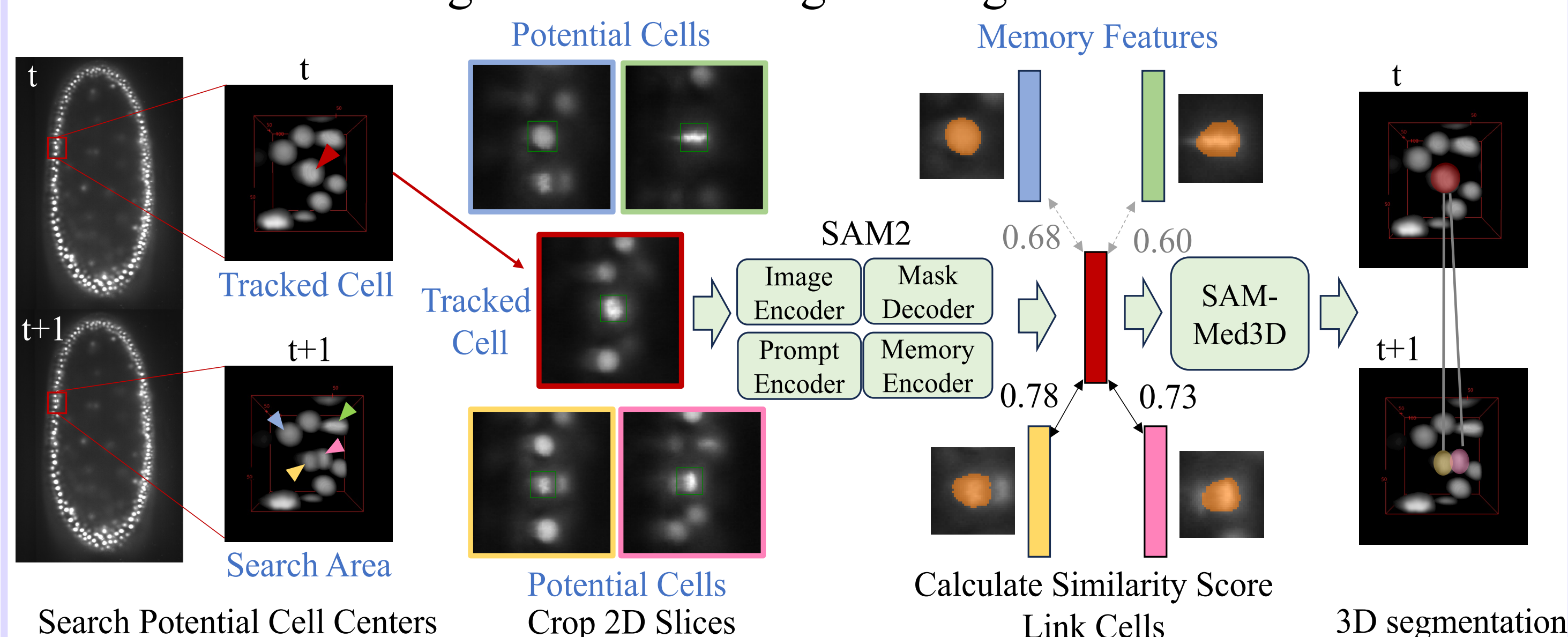
- Cell tracking: migration tracking, mitosis detection, lineage reconstruction.
- Challenges:
 - Dividing cells & mitotic events.
 - Noisy images, indistinct boundaries.
 - Dense clusters, similar appearances.
 - High annotation cost, poor generalizability across modalities.
- Our Solution:
 - Leverage foundation models (SAM2, SAM-Med3D).
 - Prompt-based local linking, followed by full lineage generation.

Method

- 2D & Small 3D Datasets (Linking only, when segmentation is given):**
 - Apply segmentation to define initial tracks and candidates for linking
 - Generate bounding box & point prompts from masks.
 - SAM2 predicts masks across consecutive frames.
 - Link masks → tracking + mitosis detection.



- Large-scale 3D+t Datasets (Segmentation & Tracking)**
 - Detect rough centers & masks (e.g. watershed).
 - Fine-tune SAM-Med3D with rough masks.
 - Forward tracking with candidate search.
 - Use memory encoder + similarity scoring for linking and mitosis detection.
 - Accurate 3D segmentation during tracking with MedSAM3D.

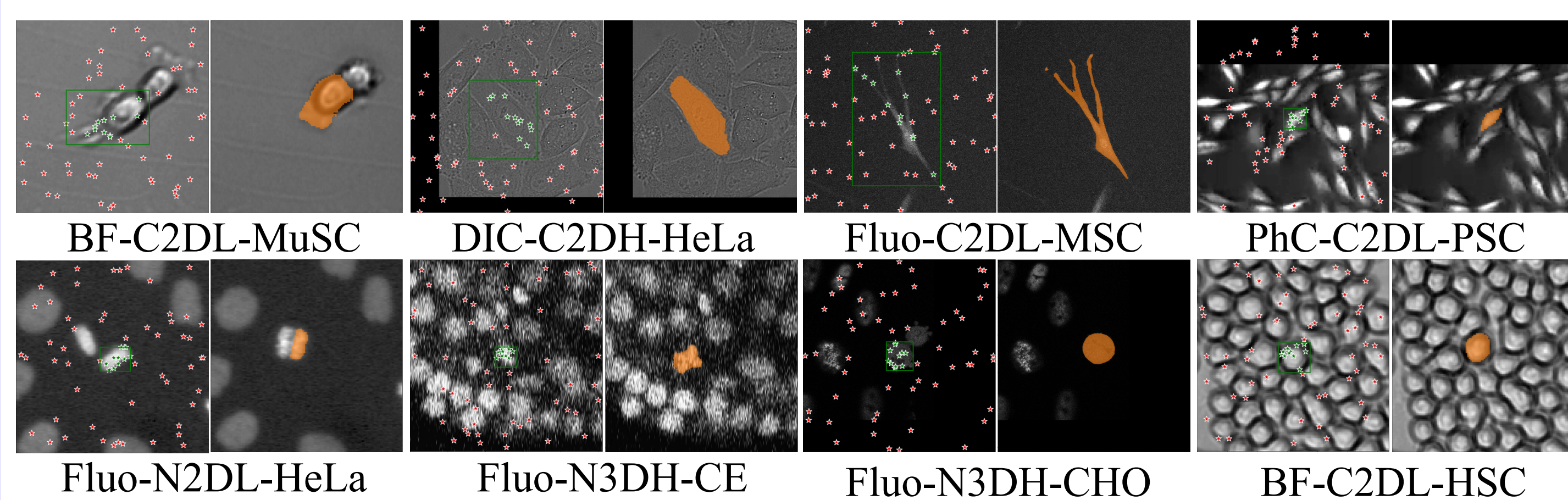


Contributions

- Zero-shot SAM-based cell tracking across frames.
- Efficient segmentation and tracking of large-scale 3D datasets.
- Fully unsupervised → no dataset-specific tuning or human annotations.
- Competitive with state-of-the-art supervised methods.
- Generalizable across different modalities and cell types.

Experimental Results

- Visualization:
 - Examples from the Cell Tracking Challenge (CTC) [3] datasets.



- Performance Highlights
 - Top 3 LNK across 13 CTC datasets.
 - 2nd TRA, 3rd SEG on large-scale 3D datasets.
 - ~1 hr for 200 frames × 100 cells (single RTX 4090).

Quantitative Results:

Cell Linking Results (ours in bold)

Dataset	All 13 Datasets	Fluo-N3DH-CE	Fluo-N3DH-CHO
LNK	1st 0.984 ⁽⁴⁾	1st 0.982 ⁽⁴⁾	1st 0.993 ⁽¹⁾
	2nd 0.977 ⁽¹⁾	2nd 0.971 ⁽¹⁾	2nd 0.990
	3rd 0.972	3rd 0.960	3rd 0.988 ⁽⁴⁾
BIO	1st 0.862 ⁽⁴⁾	1st 0.862 ⁽⁴⁾	1st 0.955 ⁽¹⁾
	2nd 0.794 ⁽¹⁾	2nd 0.782 ⁽¹⁾	2nd 0.915
	3rd 0.752 ⁽²⁾	3rd 0.667	3rd 0.890 ⁽⁵⁾
	5th 0.699		

Dataset	Fluo-N2DL-GOWT1	PhC-C2DL-PSC	Fluo-C2DL-MSC
LNK	1st 0.986 ⁽⁴⁾	1st 0.992 ⁽⁴⁾	1st 0.947 ⁽⁴⁾
	2nd 0.982	2nd 0.991 ⁽¹⁾	2nd 0.924
	3rd 0.978 ⁽¹⁾	3rd 0.991	3rd 0.921 ⁽¹⁾
BIO	1st 0.788 ⁽⁴⁾	1st 0.862 ⁽⁴⁾	1st 0.727 ⁽⁴⁾
	2nd 0.723 ⁽²⁾	2nd 0.812 ⁽¹⁾	2nd 0.657 ⁽¹⁾
	3rd 0.718 ⁽¹⁾	3rd 0.805 ⁽²⁾	3rd 0.647 ⁽³⁾
	6th 0.661	5th 0.792	6th 0.592

Cell tracking & segmentation results (ours in bold)

Dataset	Average Score	Fluo-N3DL-TRIC	Fluo-N3DL-TRIF
SEG	1st 0.714 ⁽⁶⁾	1st 0.791 ⁽⁶⁾	1st 0.746 ⁽⁹⁾
	2nd 0.700 ⁽⁹⁾	2nd 0.766 ⁽⁸⁾	2nd 0.695
	3rd 0.682	3rd 0.680 ⁽⁷⁾	3rd 0.684 ⁽¹⁰⁾
	4th 0.670 ⁽⁸⁾	4th 0.668	4th 0.654 ⁽⁷⁾
TRA	1st 0.954 ⁽⁷⁾	1st 0.952 ⁽⁷⁾	1st 0.955 ⁽⁷⁾
	2nd 0.910	2nd 0.942 ⁽⁶⁾	2nd 0.936 ⁽⁹⁾
	3rd 0.908 ⁽⁶⁾	3rd 0.899	3rd 0.921
	4th 0.895 ⁽⁹⁾	4th 0.854 ⁽⁹⁾	4th 0.886 ⁽¹⁰⁾

Indirect comparison

References

- [1] Ravi, N., Gabeur, V., Hu, Y.T., Hu, R., Ryali, C., Ma, T., Khedr, H., R.dle, R., Rolland, C., Gustafson, L., et al.: Sam 2: Segment anything in images and videos. arXiv preprint arXiv:2408.00714 (2024)
- [2] Wang, H., Guo, S., Ye, J., Deng, Z., Cheng, J., Li, T., Chen, J., Su, Y., Huang, Z., Shen, Y., et al.: Sam-med3d: towards general-purpose segmentation models for volumetric medical images. arXiv preprint arXiv:2310.15161 (2023)
- [3] Maška, M., Ulman, V., Delgado-Rodriguez, P., Gomez-de Mariscal, E., Nečasov, T., Guerrero Pea, F.A., Ren, T.I., Meyerowitz, E.M., Scherr, T., Lffler, K., et al.: The cell tracking challenge: 10 years of objective benchmarking. Nature Methods 20(7), 1010–1020 (2023)

